Imagery Data Assistant

AI-Powered Geospatial Analysis & Visualization

Version 0.1.0

Platform ArcGIS Maps SDK for JavaScript 5.0 + React

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1. Overview

The **Imagery Data Assistant** is a conversational AI-powered web application for geospatial data discovery, visualization, and analysis. Users interact with a natural-language chat interface to search for imagery and geospatial layers, load them onto a 2D or 3D map, apply raster analysis functions, take measurements, and adjust 3D elevation offsets — all without writing code or navigating complex menus.

> **Key Idea:** Instead of clicking through toolbars and dialogs, users simply describe what they want in plain English. The assistant's AI orchestrator routes each request to the appropriate specialized agent, which executes the task and reports results back in the chat.

Core Capabilities at a Glance

- **Content Search & Discovery** — Search My Content, My Org, ArcGIS Online, or Living Atlas for layers by keyword
- **Layer Loading & Navigation** — Load any ArcGIS-supported layer type from a URL, item ID, or keyword search. Navigate to places by name via geocoding (works in 2D and 3D)
- **Imagery Tools** — Apply stretches, color ramps, server-side processing templates, and identify pixel values
- **Layer Comparison** — Swipe between two layers to compare them side by side
- **Point Cloud Visualization** — Filter LiDAR by classification, color by elevation/intensity/RGB
- **Measurement Tools** — Distance, area, and elevation profile with configurable units
- **3D Elevation Offset** — Auto-fix, click-to-fix, calculate, or manually adjust floating/underground 3D layers
- **Oriented Imagery** — View street-level and oblique imagery with coverage footprints
- **Catalog Layers** — Browse and filter items in catalog layer collections
- **Layer Info** — Inspect metadata, fields, statistics, and layer order
- **Help** — Contextual help and capabilities overview

2. Architecture

```
┌──────────────────────────────────────────────────────────────────────┐
│  User Interface                                                      │
│                                                                      │
│  ┌───────────────────────┐  ┌────────────────────────────────────┐   │
│  │  arcgis-map / scene   │  │  arcgis-assistant (Chat Panel)     │   │
│  │  Map / Scene View     │  │                                    │   │
│  │  + Widgets (zoom,     │  │   User types natural-language prompt│  │
│  │    legend, layers,    │  │              │                     │   │
│  │    basemaps, measure) │  │              ▼                     │   │
│  └───────────────────────┘  │  AI Orchestrator (LLM)             │   │
│                             │   Routes to best-matching agent    │   │
│                             │              │                     │   │
│                             │              ▼                     │   │
│  Agents                     │   ┌────────────────────────────┐   │   │
│  ┌───────────────────────┐  │   │ Content Search  │ Load Layer │ │   │
│  │ Built-in:             │  │   │ Imagery Tools  │ Measurement │ │   │
│  │  • Help               │  │   │ Elev. Offset    │ Swipe      │ │   │
│  │  • Data Exploration   │  │   │ Point Cloud     │ Layer Info │ │   │
│  │    (2D only)          │  │   │ Oriented Img    │ Catalog Lyr│ │   │
│  └───────────────────────┘  │   └────────────────────────────┘   │   │
│                             └────────────────────────────────────┘   │
│                                                                      │
│  Backend Services                                                    │
│  ┌──────────────────────────────────────────────────────────────────┐│
│  │ ArcGIS Portal REST API  •  Image Server  •  Scene Server         ││
│  │ OAuth 2.0  •  Portal Search  •  Raster Functions  •  Elevation   ││
│  └──────────────────────────────────────────────────────────────────┘│
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```

How It Works

1. The user types a natural-language prompt in the **arcgis-assistant** chat panel.
2. The **AI Orchestrator** (powered by an LLM via LangChain/LangGraph) analyzes the prompt and routes it to the most relevant agent based on each agent's description.
3. The selected **agent** executes a LangGraph state graph: it extracts intent via fast regex patterns (falling back to LLM when needed), calls ArcGIS APIs, and returns a chat response.
4. Results are displayed in the chat — layers appear on the map, measurement tools activate, analysis renders, etc.

> **Smart Routing:** Each agent includes bail-out logic that detects requests meant for other agents and returns silently. This prevents conflicts when multiple agents receive the same message from the

orchestrator.

3. Technology Stack

Component	Technology
Mapping Engine	ArcGIS Maps SDK for JavaScript 5.0 (`@arcgis/core` 5.0.3)
Map Web Components	`@arcgis/map-components` 5.0.3 — `arcgis-map`, `arcgis-scene`, `arcgis-zoom`, `arcgis-home`, `arcgis-legend`, `arcgis-layer-list`, `arcgis-basemap-gallery`, measurement components
AI Assistant Components	`@arcgis/ai-components` 5.0.3 — `arcgis-assistant`, `arcgis-assistant-agent`, built-in help/navigation/data-exploration agents
UI Design System	Calcite Design System (`@esri/calcite-components` 5.0.2) — shell, panels, buttons, icons, segmented controls, loaders
AI Orchestration	LangChain LangGraph (`@langchain/langgraph` 1.1.5) — StateGraph-based agent execution with node/edge workflows
Frontend Framework	React 18.3 with TypeScript 5.9
Build Tool	Vite 7.3 with React plugin
Validation	Zod 4.3 for runtime type checking
Authentication	ArcGIS OAuth 2.0 via IdentityManager

Key ArcGIS Web Components Used

Component	Purpose
`<arcgis-map>`	2D MapView container
`<arcgis-scene>`	3D SceneView container
`<arcgis-assistant>`	AI chat panel with orchestrator
`<arcgis-assistant-agent>`	Custom agent registration element
`<arcgis-assistant-help-agent>`	Built-in help / FAQ agent

`<arcgis-assistant-navigation-agent>`	Built-in map navigation agent
`<arcgis-assistant-data-exploration-agent>`	Built-in data exploration agent
`<arcgis-distance-measurement-2d>`	2D distance measurement tool
`<arcgis-area-measurement-2d>`	2D area measurement tool
`<arcgis-direct-line-measurement-3d>`	3D line measurement tool
`<arcgis-area-measurement-3d>`	3D area measurement tool
`<arcgis-elevation-profile>`	Elevation cross-section tool
`<calcite-shell>`	Application layout shell

4. Agent Capabilities

The assistant includes **12 agents** — 2 built-in ArcGIS agents and 10 custom agents. The AI orchestrator selects the best agent for each user request based on the agent descriptions.

Help Agent BUILT-IN

Answers general questions about the assistant, its capabilities, and how to use the application. Provides contextual guidance based on the current map state.

> **Note:** The built-in `arcgis-assistant-navigation-agent` has been removed because it only supports 2D `arcgis-map` and crashes in 3D SceneView. Navigation (geocoding to places) is handled by the custom **Load Layer Agent** instead, which works in both 2D and 3D.

Data Exploration Agent BUILT-IN

Explores data attributes and statistics of layers loaded on the 2D map. Supports filtering, querying, and summarizing feature attributes.

Content Search Agent CUSTOM

Searches for geospatial layers across multiple ArcGIS sources and supports batch loading of results.

Search Scopes

- **My Content** — user's own portal items
- **My Organization** — items shared within the org
- **ArcGIS Online** — public community content
- **Living Atlas** — Esri's curated authoritative datasets
- **All** — searches all scopes simultaneously

Batch Loading

- `add result 3` — add a single result
- `add results 1-5` — add a range
- `add results 1, 3, 7` — add specific items
- `add all results` — add everything from the search

Example Prompts

- `search my content for Phoenix imagery`
- `find elevation data in Living Atlas`
- `from my org, find all layers related to wildfire`

Load Layer Agent CUSTOM

Loads any ArcGIS-supported layer type onto the map from a URL, portal item ID, or keyword search. Also handles navigation to places via geocoding (works in both 2D and 3D).

Supported Layer Types

Imagery, Feature, Tile, Map Image, Scene, Integrated Mesh, 3D Tiles, Gaussian Splat, Point Cloud, Building Scene, Oriented Imagery, WMS, WFS, WMTS, KML, GeoJSON, CSV, Vector Tile

Smart Features

- Auto-detects layer type from URL pattern or portal item metadata
- Automatically switches to 3D view when loading 3D-only layers
- Supports elevation offset on load (e.g., `load mesh with 50m offset`)
- Auto-zooms to the loaded layer's extent
- Geocodes place names using ArcGIS World Geocoding Service (replaces built-in navigation agent which only supports 2D)

Example Prompts

- `load world imagery`
- `load https://services.arcgis.com/.../ImageServer`
- `zoom to Denver CO`
- `go to Tokyo`

Imagery Tools Agent CUSTOM

Visualizes and analyzes imagery layers on the map via chat commands — stretches, color ramps, server-side processing templates, and pixel identification.

Capabilities

Category	Options
Stretch Renderers	Standard Deviation, Min-Max, Percent Clip, Histogram Equalization, Sigmoid
Color Ramps	Inferno, Viridis, Grayscale, plus all SDK named ramps (Elevation, Prediction, etc.)
Server Templates	Auto-discovers and applies server-side processing templates by name
Pixel Identify	Click-to-identify popup shows band values at any pixel; supports mosaic rules
Reset	Clear all raster functions and renderers to restore original rendering

Example Prompts

- `apply standard deviation stretch`
- `render with inferno color ramp`
- `apply the hillshade template`
- `identify pixels on click`
- `list processing templates`
- `clear raster function`

Measurement Agent CUSTOM

Activates interactive measurement tools on the map. Automatically selects the correct 2D or 3D widget based on the current view.

Measurement Types

Type	2D Component	3D Component
Distance	`arcgis-distance-measurement-2d`	`arcgis-direct-line-measurement-3d`
Area	`arcgis-area-measurement-2d`	`arcgis-area-measurement-3d`
Elevation Profile	`arcgis-elevation-profile` (works in both)	

Supported Units

- **Linear:** meters, kilometers, feet, miles, yards, nautical miles
- **Area:** square meters, square kilometers, square feet, square miles, acres, hectares

Example Prompts

- `measure distance in kilometers`
- `measure area in acres`
- `show elevation profile`
- `clear measurement`

Elevation Offset Agent CUSTOM

Fixes 3D layers that appear floating above or sunken below the terrain by adjusting their elevation offset. Supports automatic terrain-based correction and manual adjustments.

Modes

- **Auto-fix** — Samples terrain at a 5×5 grid, uses hitTest to detect the layer surface, and computes the exact delta to snap the layer to the ground
- **Click-to-fix** — Click directly on the 3D layer to snap it to terrain at that point; uses hitTest to measure the gap between layer surface and ground
- **Calculate Offset** — Diagnostic mode that measures the gap without applying changes; reports how far above or below the ground the layer is
- **Manual Set** — Sets an absolute offset value (e.g., `set offset to 50 meters`)
- **Relative Adjust** — Adds or subtracts from current offset (e.g., `raise by 10m`)
- **Check** — Reports current elevation info and Z-range without making changes

Supported 3D Layer Types

Integrated Mesh, Scene Layer, Point Cloud, Building Scene Layer, Gaussian Splat, 3D Tiles

Example Prompts

- `fix the floating mesh`
- `fix elevation by clicking`
- `calculate offset`
- `raise by 2 meters`
- `lower the mesh by 20 feet`

Swipe / Compare Agent CUSTOM

Activates a swipe tool to visually compare two layers side by side on the map. Drag a handle across the map to reveal one layer on each side.

Features

- Horizontal (left/right) and vertical (top/bottom) swipe directions
- Accepts layer names or numeric references (e.g., `compare layer 1 and layer 3`)
- Swipe position defaults to center; can be customized

Example Prompts

- `compare the two imagery layers`
- `swipe between NAIP and Sentinel`
- `clear swipe`

Point Cloud Agent CUSTOM

Controls point cloud layer visualization — classification filtering, color modes, point sizing, and density for LiDAR data.

Capabilities

- **Classification Filter** — Show/hide by class code (ground, vegetation, buildings, water, etc.)
- **Color By** — Elevation, intensity, classification, RGB, or return number
- **Point Size & Density** — Adjust visual point size and points-per-inch

Example Prompts

- `show only ground and buildings`
- `color by elevation`
- `increase point size`

Oriented Imagery Agent CUSTOM

Opens the ArcGIS Oriented Imagery viewer for street-level, oblique, or drone imagery layers. Supports coverage footprints, image galleries, and navigation tools.

Example Prompts

- `open oriented imagery viewer`
- `show coverage footprints`
- `close viewer`

Catalog Layer Agent CUSTOM

Manages Catalog Layers (dynamic collections of items organized by geography). Provides an interactive filter panel for browsing items by type within the catalog.

Example Prompts

- `show the catalog filter panel`
- `filter catalog by imagery`

Layer Info Agent CUSTOM

Queries layers for detailed metadata, fields, attributes, popup configuration, capabilities, spatial reference, processing templates, and raster statistics.

Features

- Supports numeric references: `describe layer 4`
- Lists layer order / draw order
- Computes raster statistics (min, max, mean, std dev) for imagery and elevation layers
- Reports sublayers, tables, band count, pixel type, and spatial reference

Example Prompts

- `describe the imagery layer`
- `what is the layer order?`
- `show fields for layer 2`
- `what are the raster statistics?`

5. Example Workflows

Workflow 1: Discover and Analyze Imagery

Scenario: A GIS analyst wants to find and analyze thermal imagery for a city.

1. `search Living Atlas for thermal imagery in Phoenix`
   Content Search Agent returns a numbered list of matching imagery layers.
2. `add result 1`
   The layer is loaded onto the map and the view zooms to its extent.
3. `apply standard deviation stretch`
   Imagery Tools Agent enhances the contrast for better visualization.
4. `render with inferno color ramp`
   A heat-style color ramp is applied to highlight temperature differences.
5. `identify pixels on click`
   Clicking anywhere on the imagery now shows pixel/band values.
6. `list processing templates`
   Shows all server-side raster functions available for this layer.

Workflow 2: Batch Load Organizational Content

Scenario: A project manager wants to load all team layers for a project area.

1. `from my content, find all layers related to Phoenix`
   Content Search Agent searches the user's portal content and returns all matching items.
2. `add all results`
   All results are loaded concurrently onto the map. If any require 3D, the view auto-switches to a Scene before loading.
3. `measure area in acres`
   The area measurement tool activates for the user to draw a polygon.

Workflow 3: Load and Fix a 3D Mesh

Scenario: A drone survey produced a 3D mesh that floats above the terrain.

1. `load https://tiles.arcgis.com/ … /3DTilesServer/tileset.json`
   Load Layer Agent detects a 3D Tiles layer, auto-switches to 3D SceneView, and loads it.
2. `fix the floating mesh`
   Elevation Offset Agent uses hitTest to detect the layer surface, samples terrain, and snaps the layer to the ground automatically.
3. `calculate offset`
   Reports exactly how far the layer is from the ground without changing anything.
4. `raise by 2 meters`
   Fine-tunes the offset with a relative +2m adjustment.
5. `measure distance in feet`
   3D direct-line measurement tool activates in feet for verification.
6. `elevation profile`
   The elevation profile tool lets the user draw a line across the mesh to verify alignment.

Workflow 4: Multi-Source Comparison

Scenario: Comparing elevation data from different sources.

1. `search ArcGIS Online for NAIP imagery Arizona`
   Finds NAIP aerial imagery layers.
2. `add result 1`
   Loads the NAIP imagery as a base reference layer.
3. `apply NDVI`
   Applies Normalized Difference Vegetation Index to visualize vegetation health.
4. `find elevation data in Living Atlas`
   Searches for DEM / terrain layers in Esri's curated collection.
5. `add results 1-3`
   Loads the top three elevation datasets for comparison.
6. `apply hillshade to the elevation layer`
   Renders terrain shading for better visual interpretation.

Workflow 5: Quick Q&A and Exploration

Scenario: A new user learning the application.

1. `what can you do?`
   Help Agent explains the assistant's capabilities and available commands.
2. `zoom to San Francisco`
   Navigation Agent flies the map to San Francisco.
3. `load world imagery`
   Loads the Esri World Imagery basemap layer.
4. `how far is it from here to Oakland?`
   Measurement Agent activates the distance tool for the user to measure.

6. Getting Started

Prerequisites

- Node.js 18+ and npm
- An ArcGIS Online or ArcGIS Enterprise account
- An OAuth 2.0 Application ID registered in ArcGIS (for portal content access)

Setup

Environment Variables Required:
`VITE_ARCGIS_OAUTH_APP_ID` — Your ArcGIS OAuth application ID (required)
`VITE_ARCGIS_PORTAL_URL` — Portal URL (optional, defaults to https://www.arcgis.com)
`VITE_WEBMAP_ITEM_ID` — Existing web map ID (optional, auto-creates if not set)

1. Clone the repository and navigate to `Apps/imagery-assistant`
2. Create a `.env` file with your OAuth App ID
3. Install dependencies: `npm install`
4. Start the dev server: `npm run dev`
5. Open the URL shown in your browser and sign in with your ArcGIS account

Build for Production

Run `npm run build` to produce an optimized production build in the `dist/` folder.

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